

MODULE 04

Modern .NET API and Data Access

Day 1 · Core MVP Module



ADOBE STOCK PLACEHOLDER

Suggested search: "REST API code on a monitor (wide hero banner)"

LEARNING OBJECTIVES

- Build a modern API endpoint using current .NET patterns.
- Integrate data access through EF Core or Dapper.
- Implement robust validation and configuration handling.



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Suggested search: "database server room (supporting image)"

WHY THIS MODULE MATTERS

This module creates the first meaningful modernization increment.

It delivers a production-relevant API capability with data persistence and test coverage.

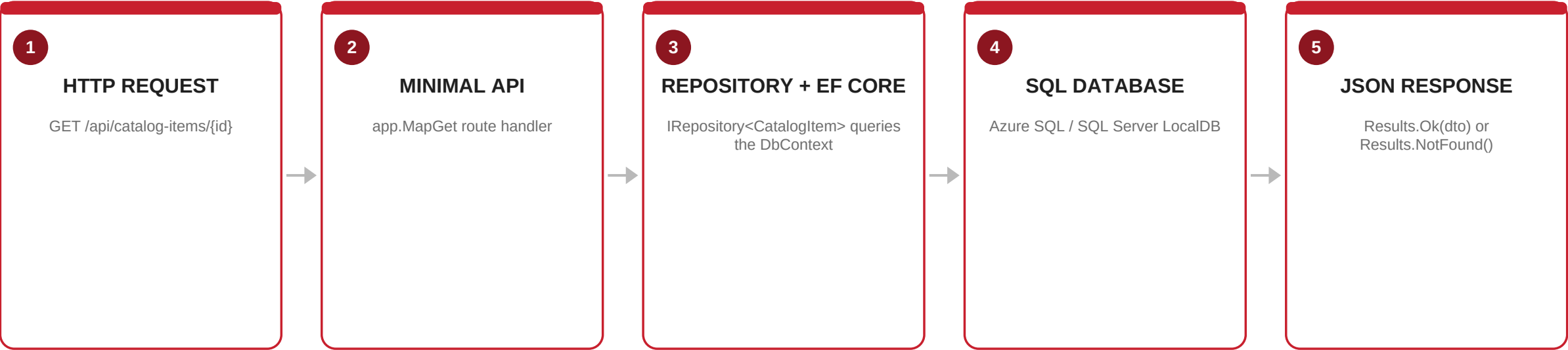


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Suggested search: "developer building a web API (concept image)"

CONCEPT MAP

The exact request path a Minimal API endpoint follows in eShopOnWeb's PublicApi project.



FROM THE DOCS: WHY MINIMAL APIS

- Microsoft recommends Minimal APIs over controller-based APIs for new ASP.NET Core projects.
- Minimal APIs build working REST endpoints with minimal code — no controller classes or scaffolding.
- Endpoints are defined with `app.MapGet()`, `MapPost()`, `MapPut()`, `MapDelete()` on the `WebApplication` instance.
- Route handlers can request services directly — ASP.NET Core resolves them from DI automatically.
- Dapper is a lightweight object mapper for raw SQL; EF Core is a full ORM with change tracking and migrations.



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Suggested search: "software architecture data flow diagram (detail image)"

COURSEWARE FOCUS

- Minimal APIs
- Dependency injection
- Configuration
- Async/await
- Data access



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REAL ENDPOINT — ESHOPONWEB PUBLICAPI

REAL CODE — src/PublicApi/CatalogItemEndpoints/CatalogItemGetByIdEndpoint.cs

```
public class CatalogItemGetByIdEndpoint
    : IEndpoint<IResult, GetByIdCatalogItemRequest, IRepository<CatalogItem>>
{
    public void AddRoute(IEndpointRouteBuilder app)
    {
        app.MapGet("api/catalog-items/{catalogItemId}",
            async (int catalogItemId, IRepository<CatalogItem> itemRepository) =>
            {
                return await HandleAsync(
                    new GetByIdCatalogItemRequest(catalogItemId), itemRepository);
            })
            .Produces<GetByIdCatalogItemResponse>()
            .WithTags("CatalogItemEndpoints");
    }

    // HandleAsync queries the repository, then returns
    // Results.NotFound() or Results.Ok(response)
}
```

This is the exact Minimal API + repository pattern Lab 04 extends with a new endpoint.

TOOLS & RESOURCES

Sample Code

.NET 8 / .NET 9 Web API Reference Samples (dotnet/samples)

Template for Minimal APIs, Dependency Injection, and Async/Await.

ORM / Database

Entity Framework Core & Dapper docs and code snippets

Adding REST endpoints backed by Azure SQL / SQL Server LocalDB.

Security Guides

Azure Key Vault references for ASP.NET Core

Secrets management lab exercise.

SUPPORTING ASSETS

- App baseline: [course/repos/eShopOnWeb](#)
- Pattern references: [course/repos/samples](#)



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Suggested search: "database server room (reference screenshot)"

LAB 04: MODERN .NET API + SQL



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Suggested search: "developer building a web API (lab / hands-on photo)"

LAB STEPS

Goal: Add a SQL-backed API endpoint and validate it end to end.

Tools: .NET SDK, SQL tooling · Cloud: optional Azure SQL, otherwise local SQL.

- Step 1 — Add a new API endpoint for a domain entity.
- Step 2 — Register services via dependency injection.
- Step 3 — Add EF Core or Dapper data access logic.
- Step 4 — Add configuration and secure secret handling.
- Step 5 — Add and run integration tests.
- Validation: the endpoint returns expected codes/schema, and integration tests pass.



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Suggested search: "software architecture data flow diagram (hands-on lab photo)"

KNOWLEDGE CHECK: Modern .NET API and Data Access

Which two data-access approaches are named for connecting a modern .NET API to a database in this module?

- A. EF Core and Dapper
- B. jQuery and AJAX
- C. XML and SOAP
- D. FTP and SCP



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KNOWLEDGE CHECK: Modern .NET API and Data Access

Per Microsoft's own ASP.NET Core docs, which API style is now recommended for new projects?

- A. SOAP services
- B. Minimal APIs
- C. WCF
- D. Controller-based APIs only



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SOURCES & FURTHER READING

EXTERNAL DOCUMENTATION

■ APIs overview — ASP.NET Core

<https://learn.microsoft.com/en-us/aspnet/core/fundamentals/apis>

REFERENCED IN THIS REPOSITORY

- `course/repos/eShopOnWeb/src/PublicApi/CatalogItemEndpoints/CatalogItemGetByIdEndpoint.cs`
- `course/mvp-delivery/labs/lab-04-modern-dotnet-api-sql.md`

RECAP & SUCCESS CRITERIA

- Endpoint passes functional and integration checks and follows secure config handling.
- Tier: Core MVP Module



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Suggested search: "developer building a web API (team success photo)"